

Welcome

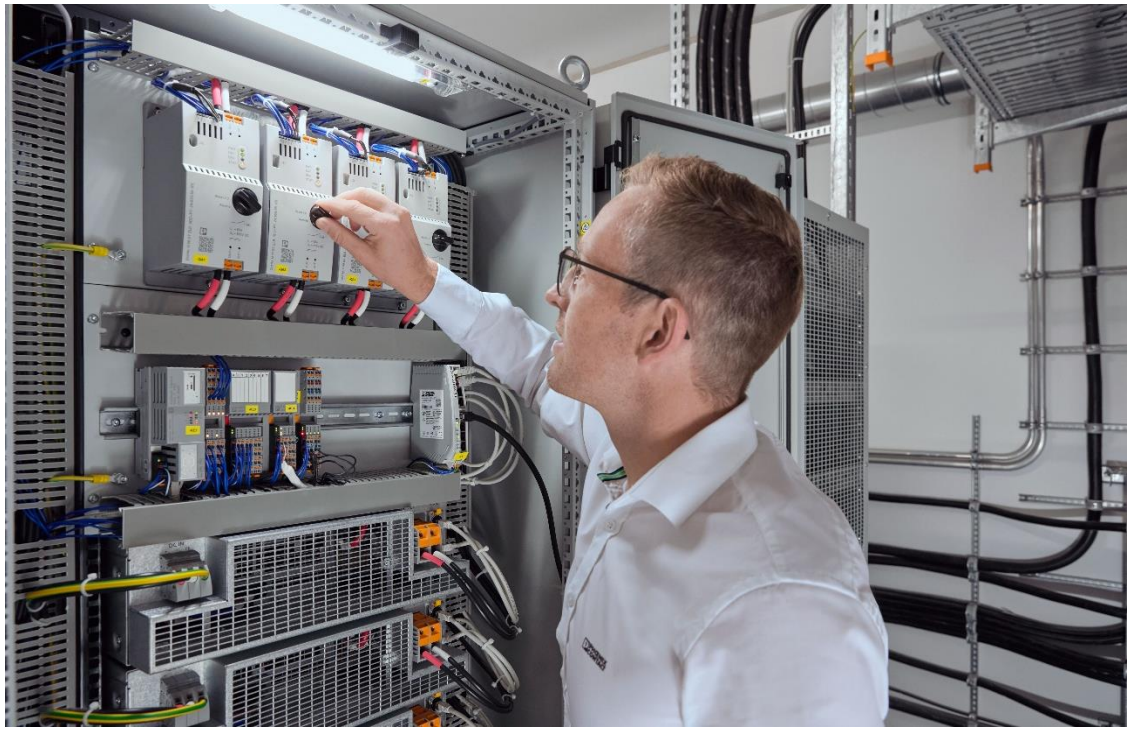
Scaling Direct Current in Data Centers

- From Industrial Proven Concepts to Open Standardization

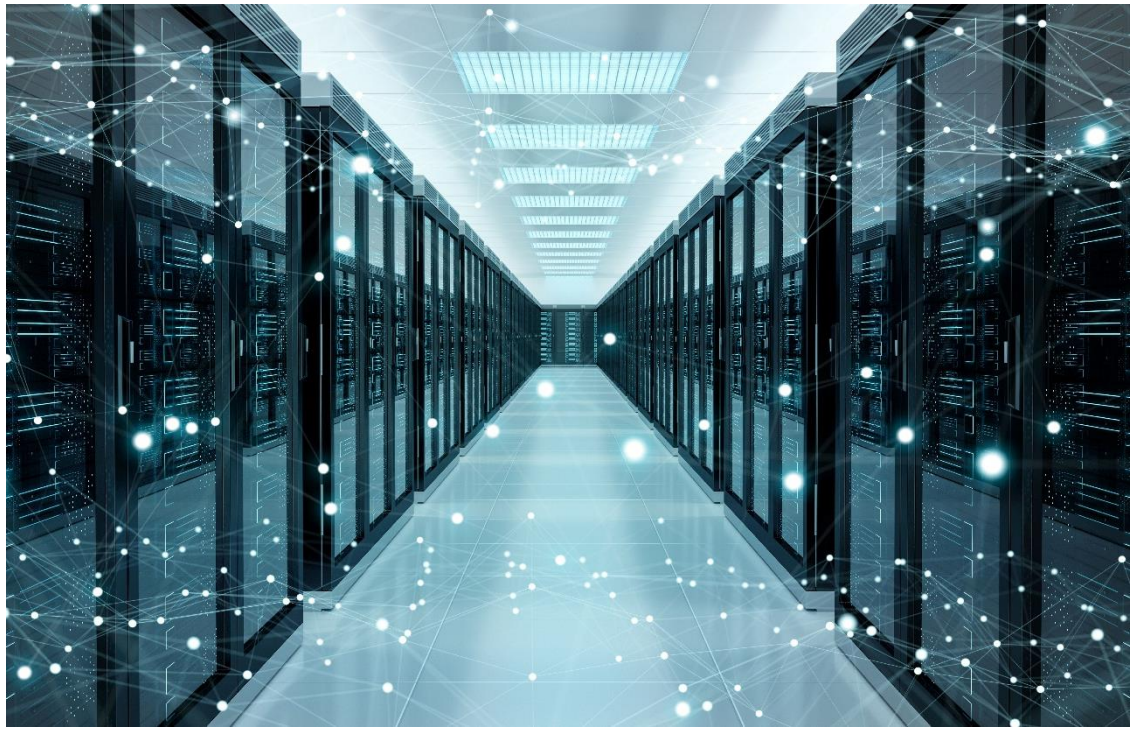
Dr. Martin Wetter
Executive Vice President Power Reliability
Phoenix Contact GmbH & Co. KG, Germany

My Motivation for this Speech is..

.. to share our experience with setting up and operating an industrial DC grid ..



.. so that these insights can be used for future architectures in data centers!



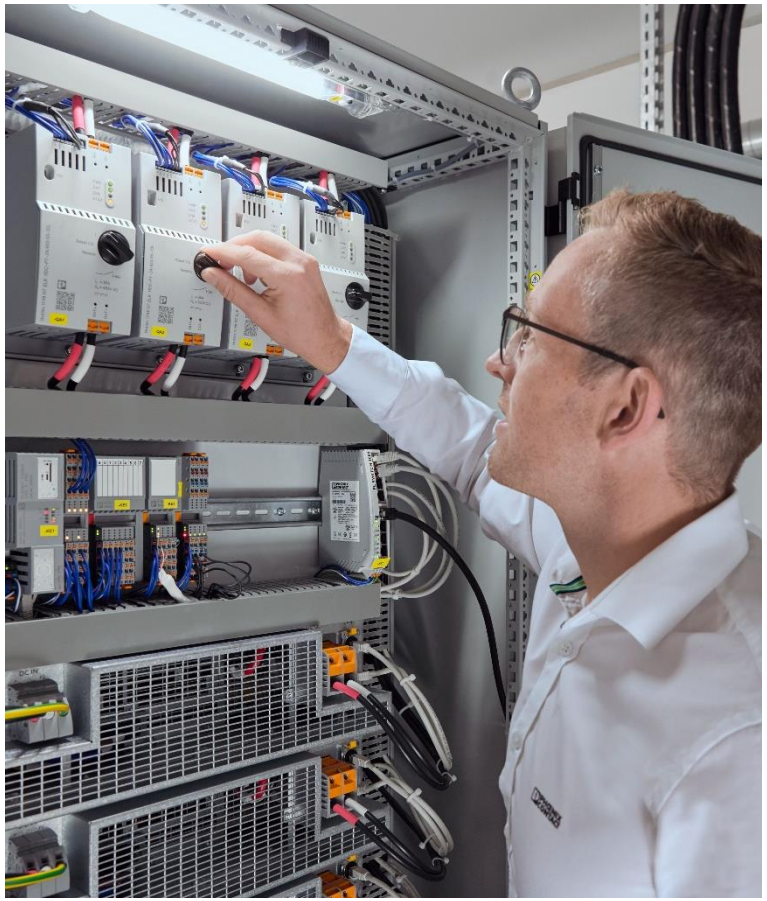
Agenda

Industrial Application

- Why using DC Grids in Industrial Applications?
- Setting up a DC grid for industrial applications
  - Topology
  - Challenges and Solutions

Data Center

- Why using DC Grids in Data Centers?
- Speeding up the Implementation



A Decade of State-Funded Research Projects on DC for Industry

	Project	Duration
1	AREUS	2013-2016
2	DC4Cities	2013-2016
3	DC INDUSTRIE	2016-2019
4	DC Schutzorgane	2016-2019
5	DCNextEve	2017-2022
6	RDC2MT	2017-2022
7	R2D2-NET	2018-2020
8	DC INDUSTRIE 2	2019-2023
9	DC Schutzsystem	2019-2023
10	EMPOWER	2019-2024
11	FASS	2020-2023
12	HYPERRIDE	2020-2025
13	**HybScha DC**	**2023-2026**
14	**DCI4CHARGE**	**2023-2026**
15	**SHIFT2DC**	**2023-2027**
16	**DC LabSaxony**	**2024-2031**
17	**DC-POWER**	**2024-2027**
18	**FEN ModuL**	**2025-2029**

DC-INDUSTRIE
ENERGIEWENDE TRIFFT INDUSTRIE 4.0

DC CAMPUS

Gefördert durch:
Bundesministerium für Wirtschaft und Klimaschutz
aufgrund eines Beschlusses des Deutschen Bundestages

SEVENTH FRAMEWORK PROGRAMME

HORIZON EUROPE

PHOENIX CONTACT

Why Using DC Grids in Industrial Applications?

Energy efficiency

Reduce conversion losses

Peak shaving

Reduce load peaks & use breaking energy

Grid stability

Easily integrate energy storage and decrease dependency AC grid

Less resources

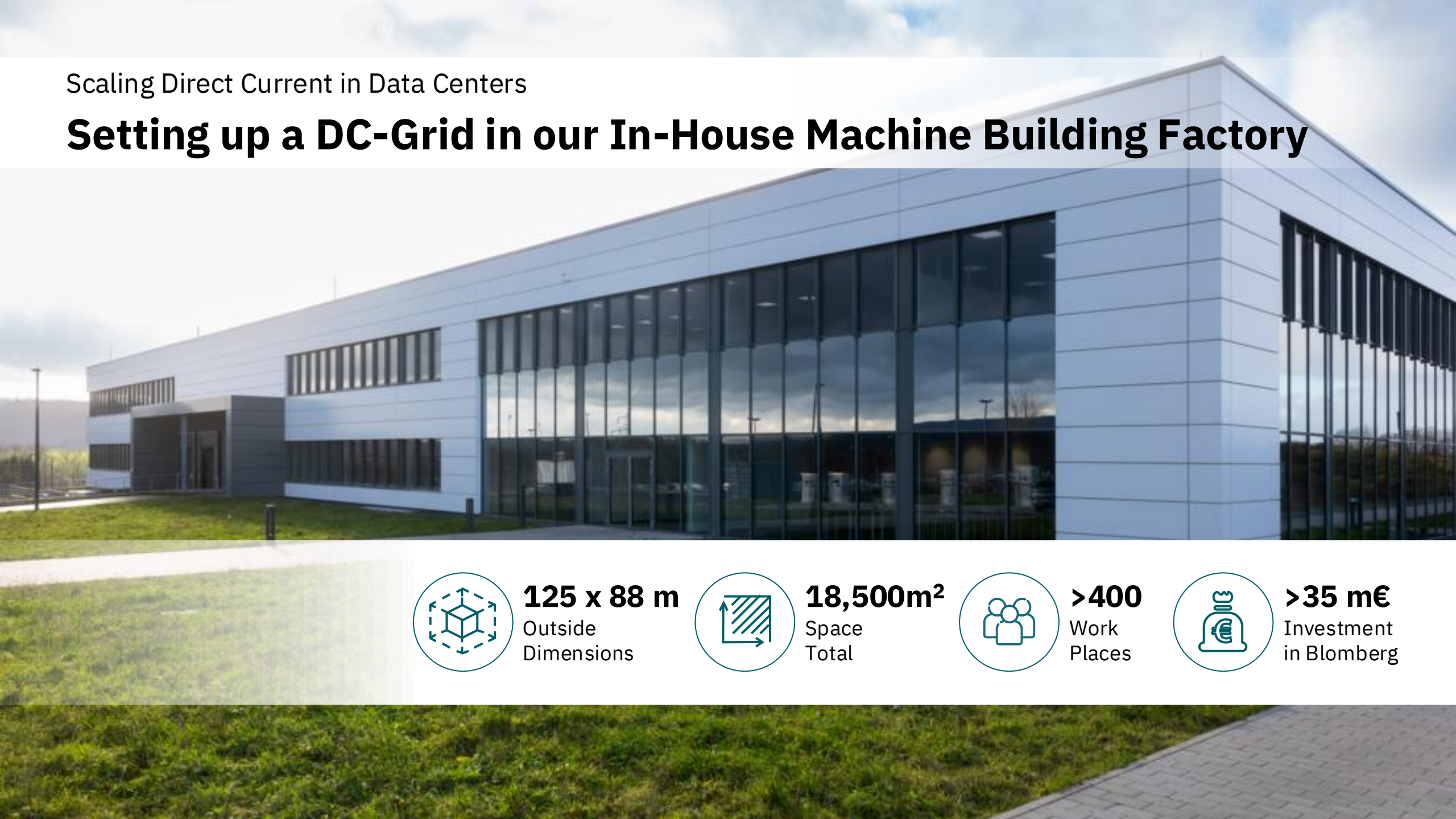
Use less copper and smaller components

Sustainability

Easily integrate renewables

Scaling Direct Current in Data Centers

Setting up a DC-Grid in our In-House Machine Building Factory



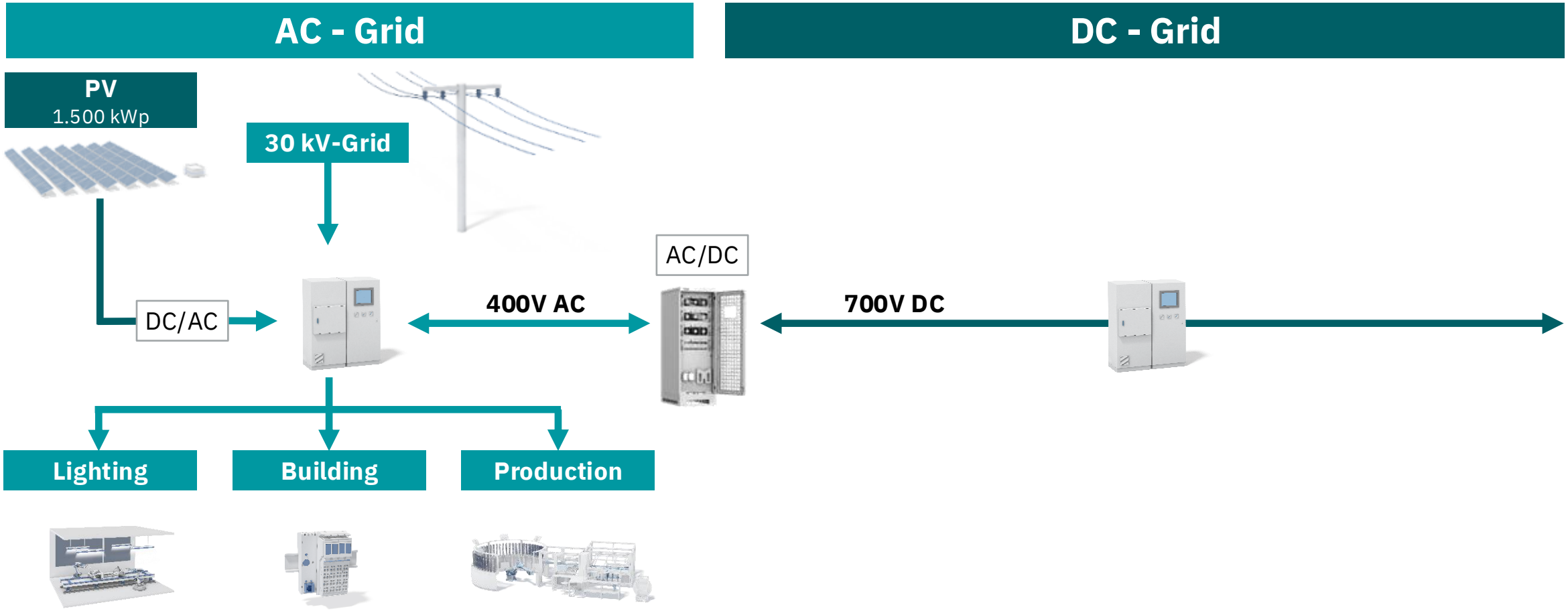
125 x 88 m
Outside Dimensions

18,500m²
Space Total

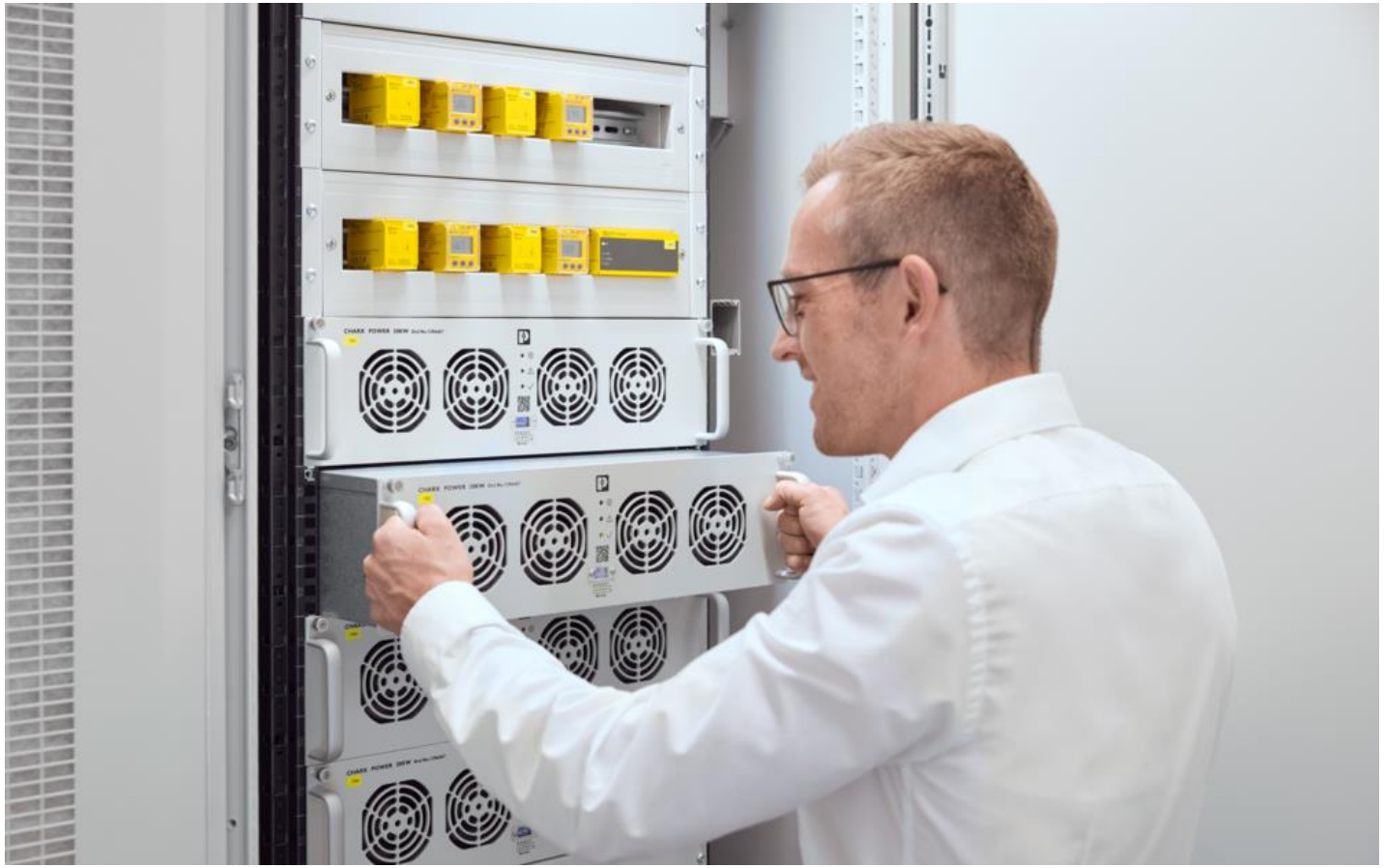
>400
Work Places

>35 m€
Investment in Blomberg

Topology of the DC-Grid in our In-House Machine Building Factory

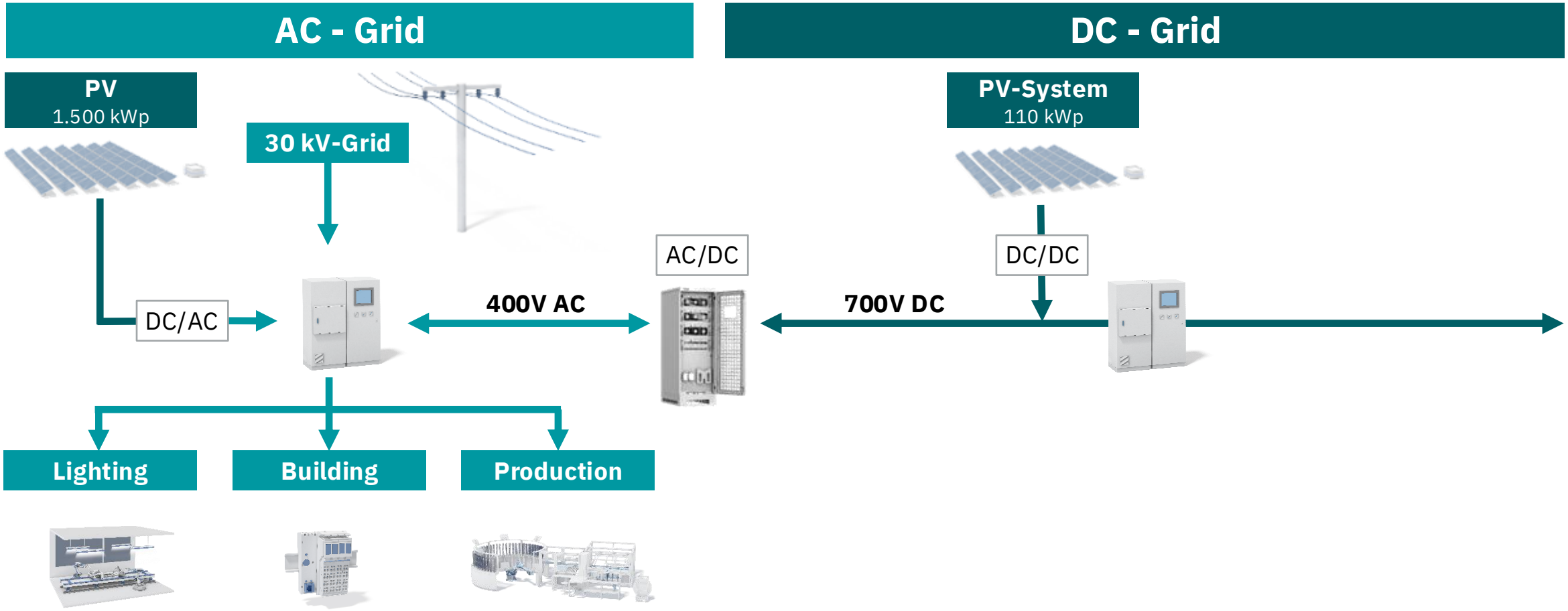


Bidirectional AC/DC converter – 700 kW AIC





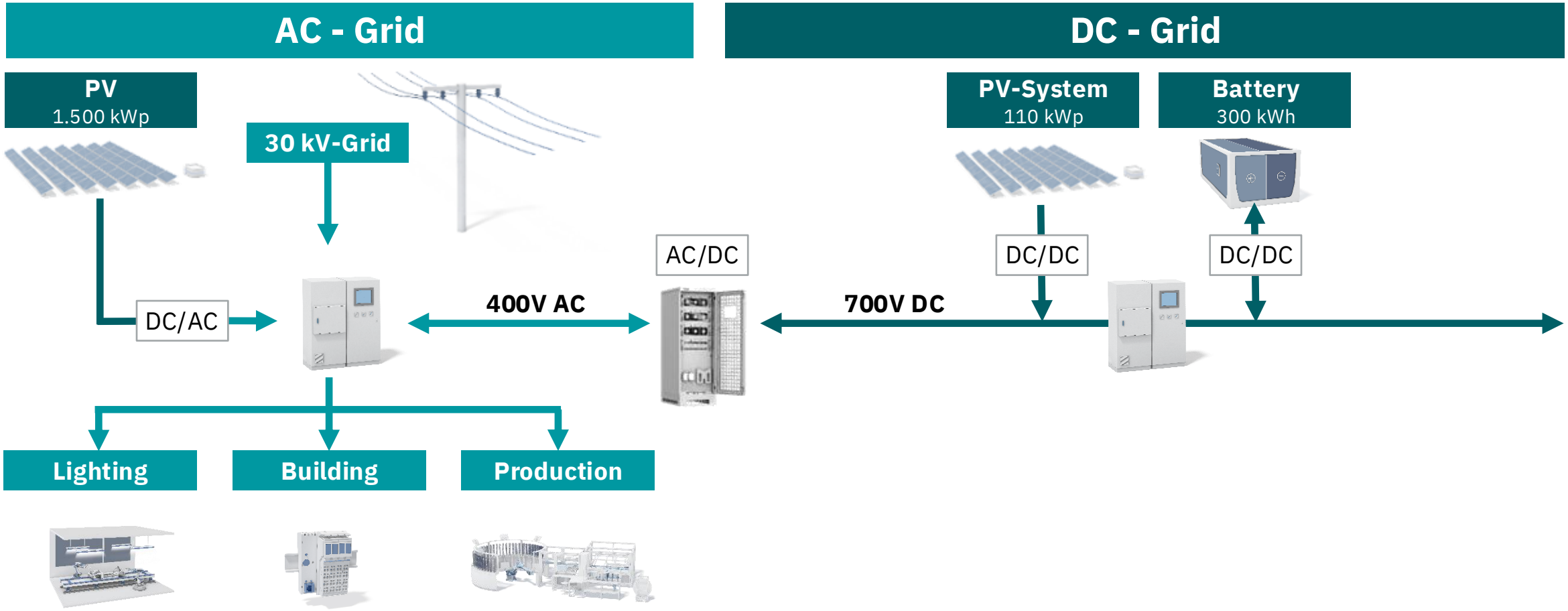
Topology of the DC-Grid in our In-House Machine Building Factory



PV-System: 1.500 kWp DC/AC + 110 kWp DC/DC

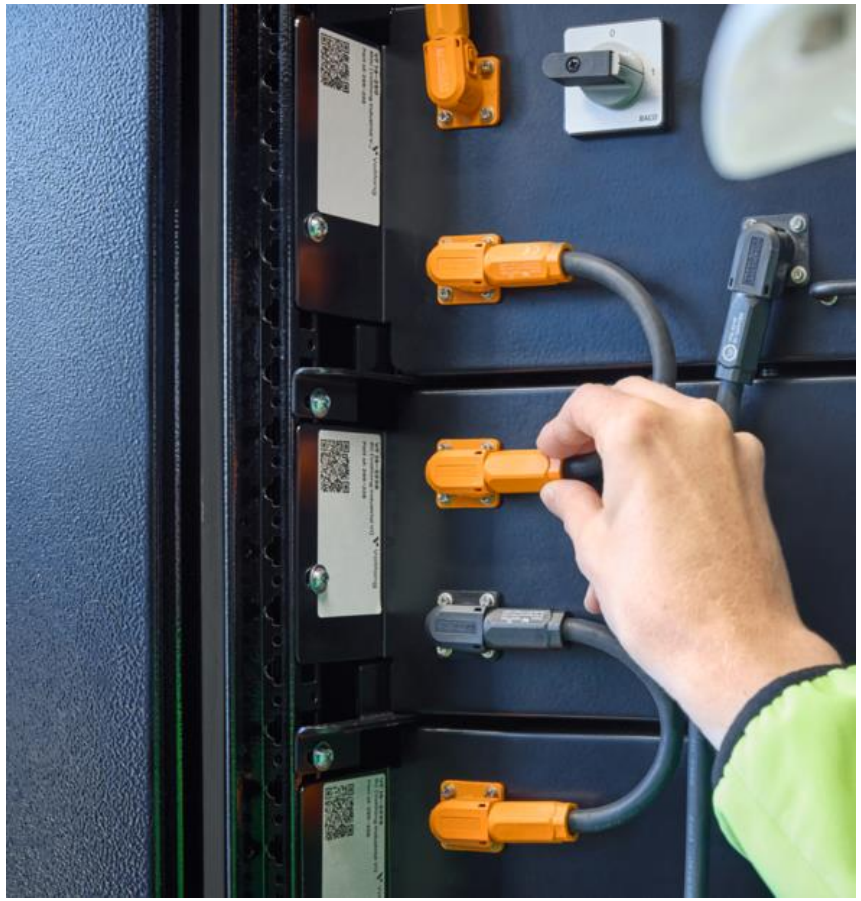


Topology of the DC-Grid in our In-House Machine Building Factory

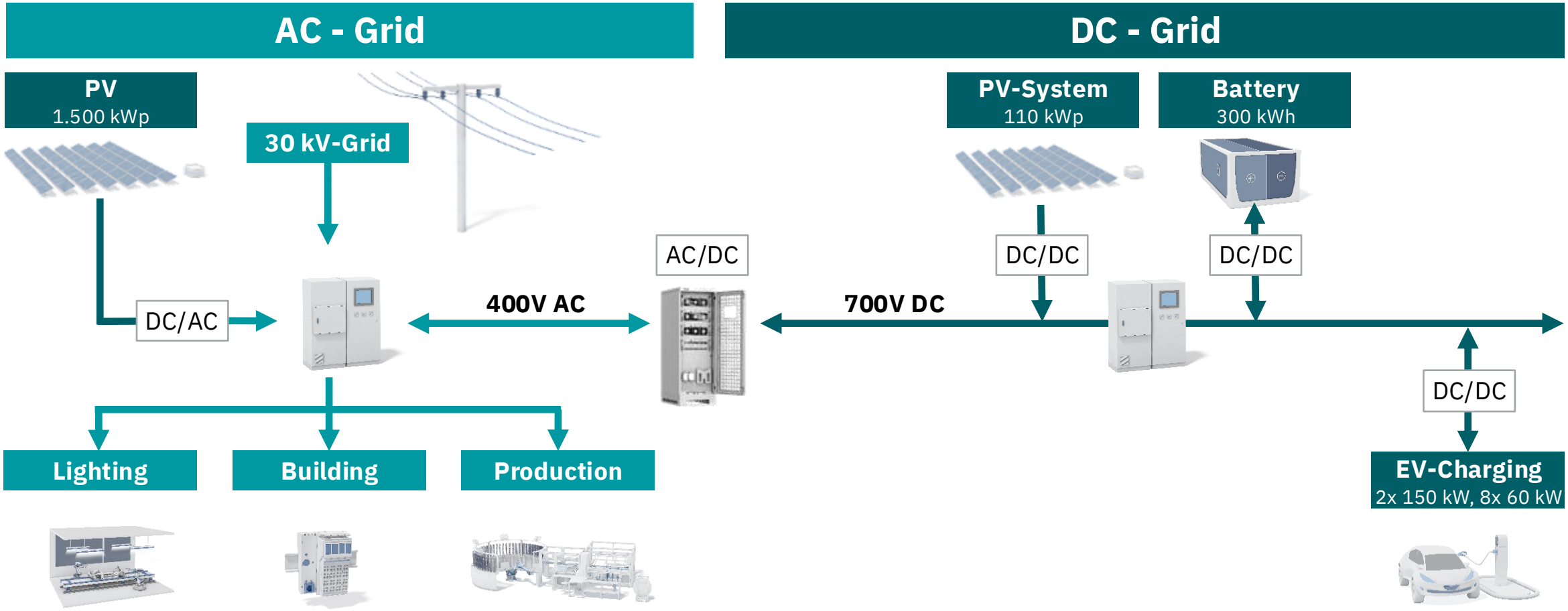


Battery Storage System 300 kWh





Topology of the DC-Grid in our In-House Machine Building Factory

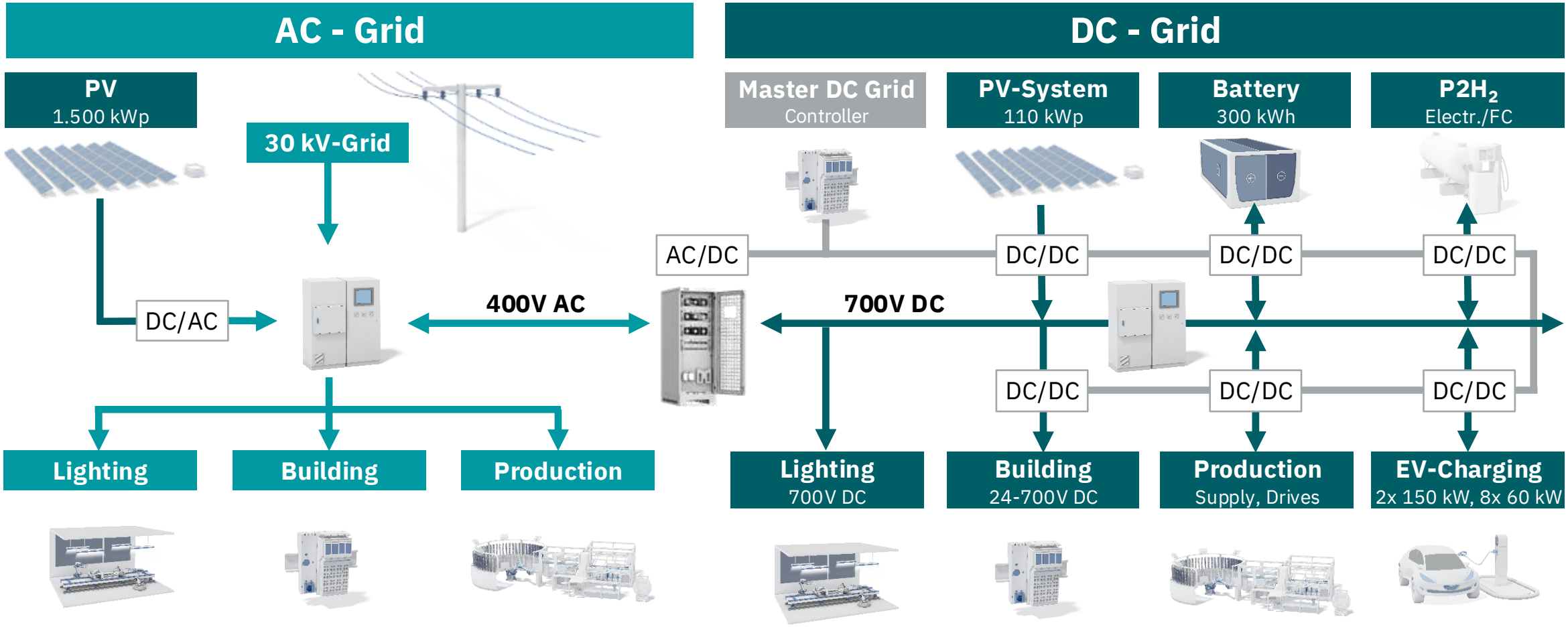


Bidirectional EV Charging: 8 x 60 kW + 2 x 150 kW



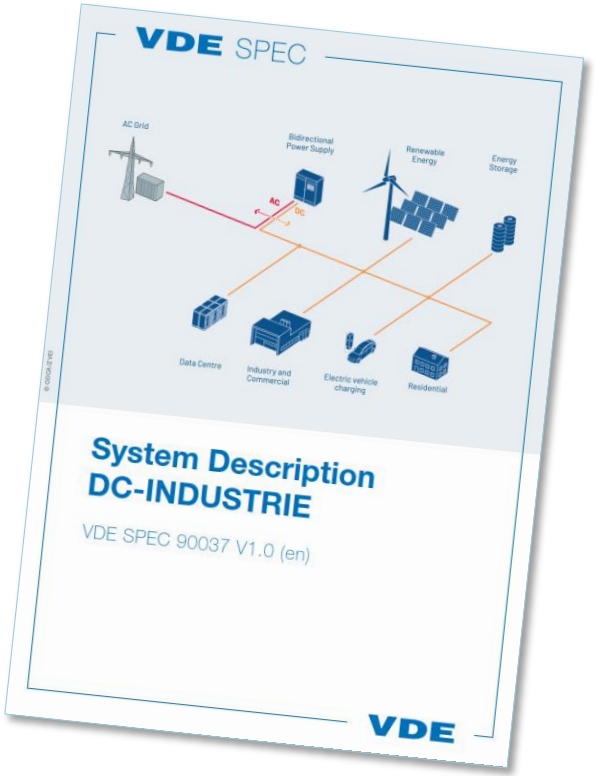


Topology of the DC-Grid in our In-House Machine Building Factory

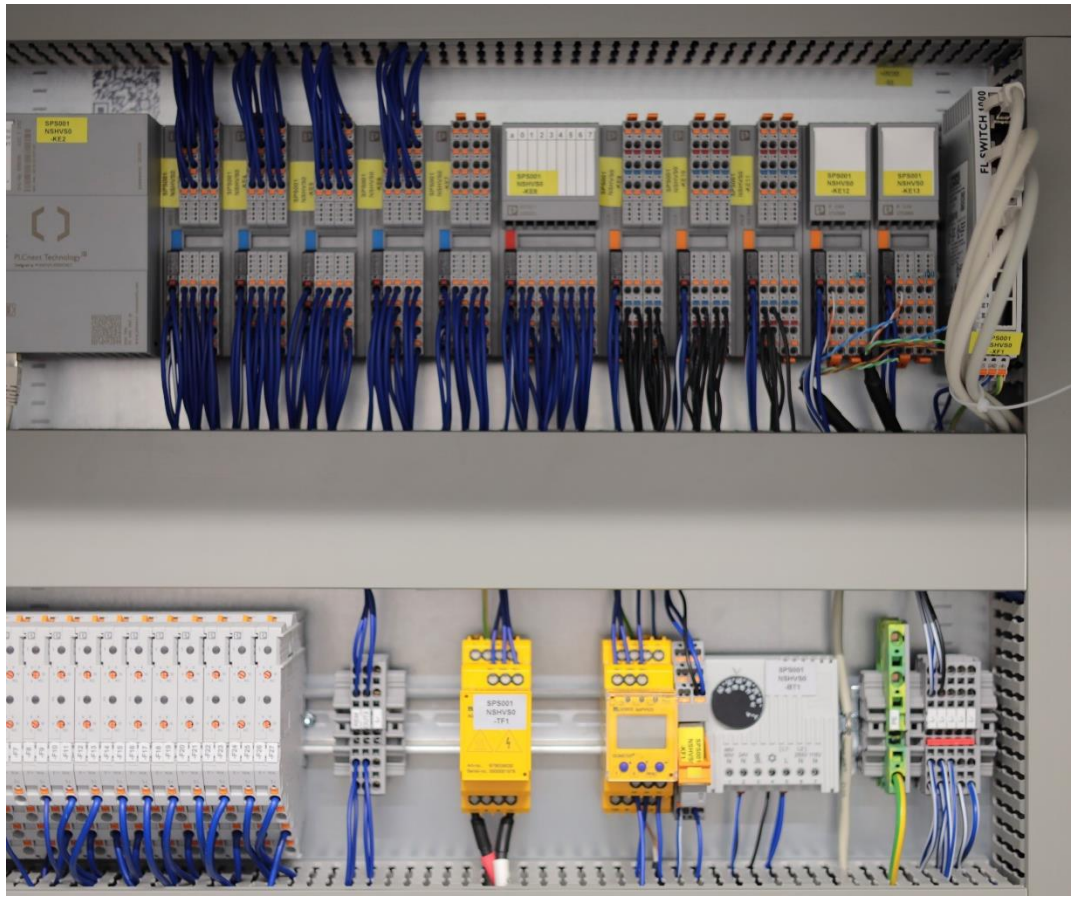


Grid Control for stable operation – DC Grid

Grid control by voltage



ODCA direct current by zvei



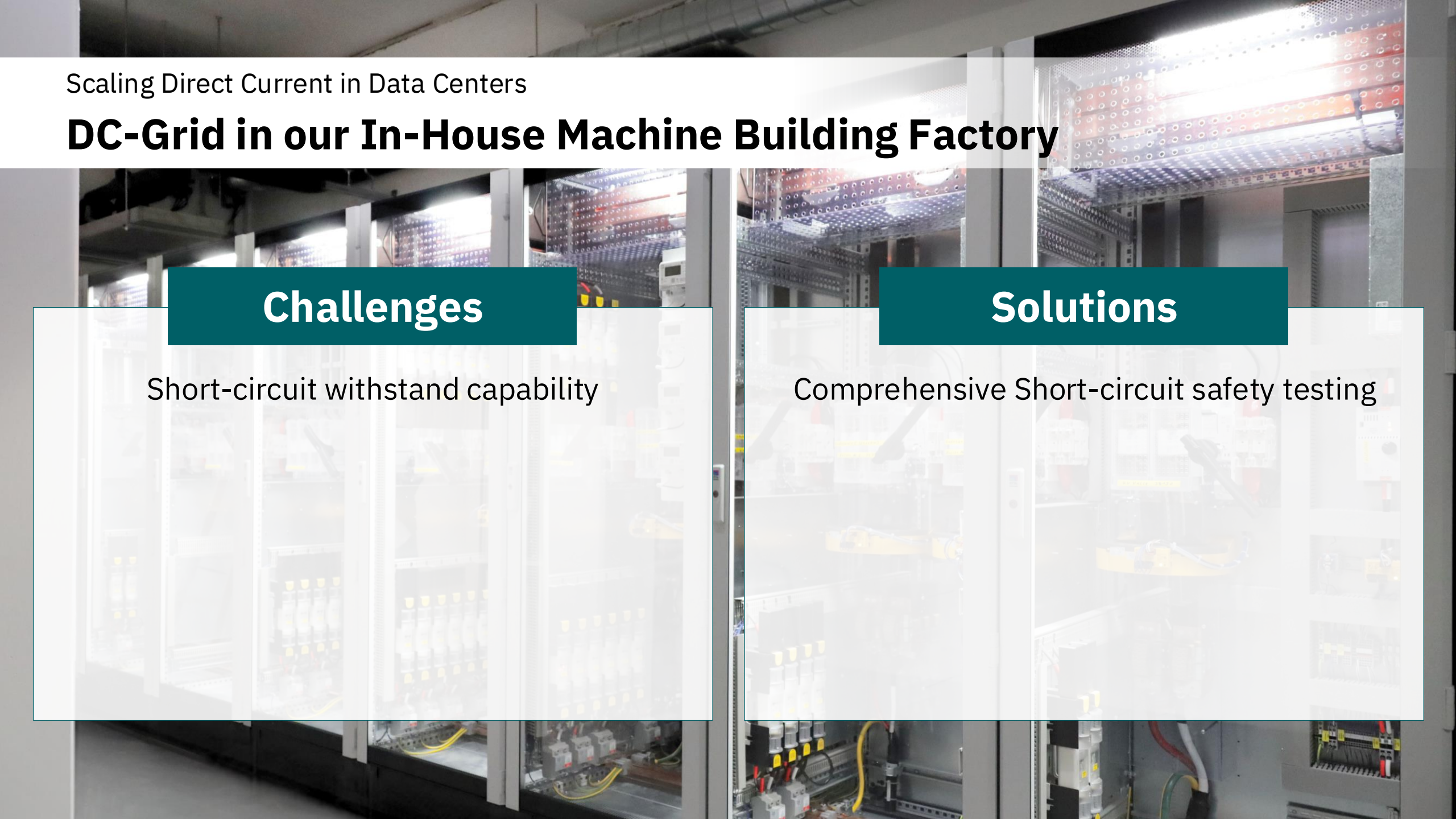
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DC-Grid in our In-House Machine Building Factory



Scaling Direct Current in Data Centers

DC-Grid in our In-House Machine Building Factory



Challenges

Short-circuit withstand capability

Solutions

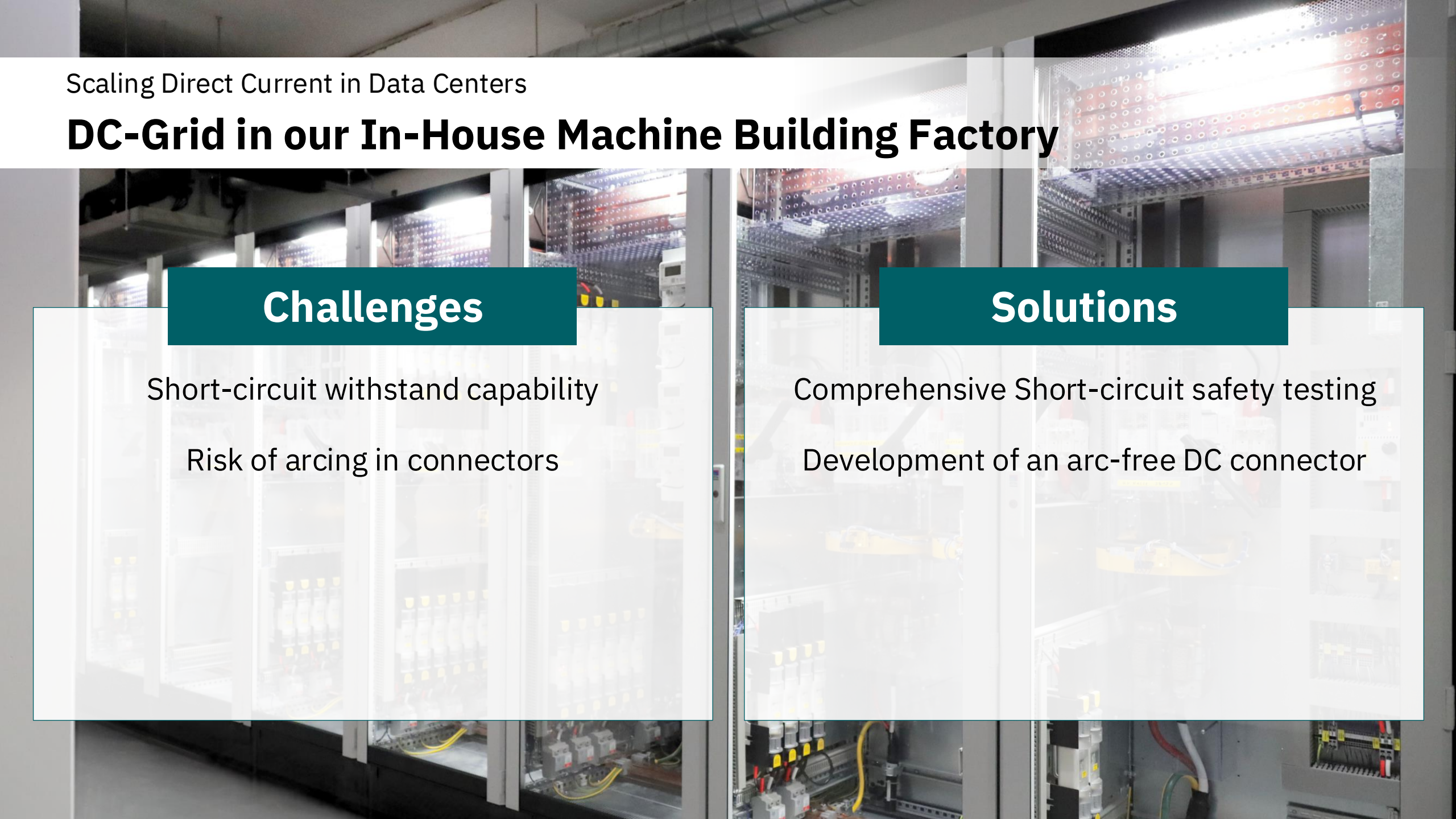
Comprehensive Short-circuit safety testing

Short-circuit Safety Testing (Cabinets with Viewing Doors)



Scaling Direct Current in Data Centers

DC-Grid in our In-House Machine Building Factory



Challenges

Short-circuit withstand capability

Risk of arcing in connectors

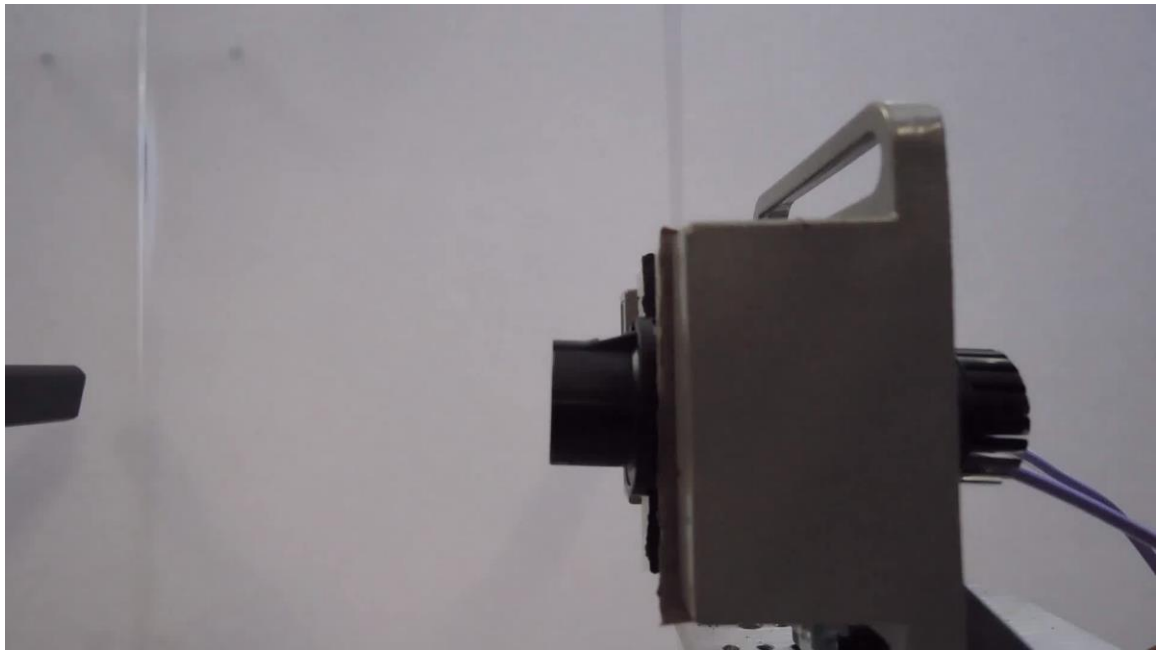
Solutions

Comprehensive Short-circuit safety testing

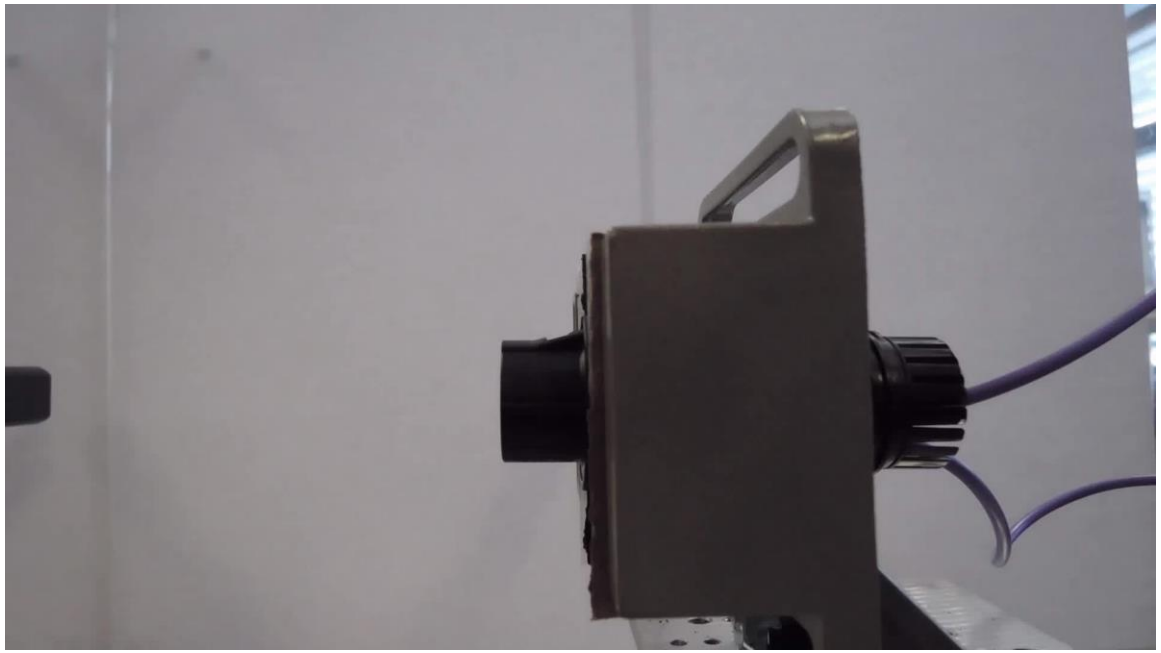
Development of an arc-free DC connector

Disconnecting a Standard AC Connector (320 V, 20A)

AC Load: 320V, 20A



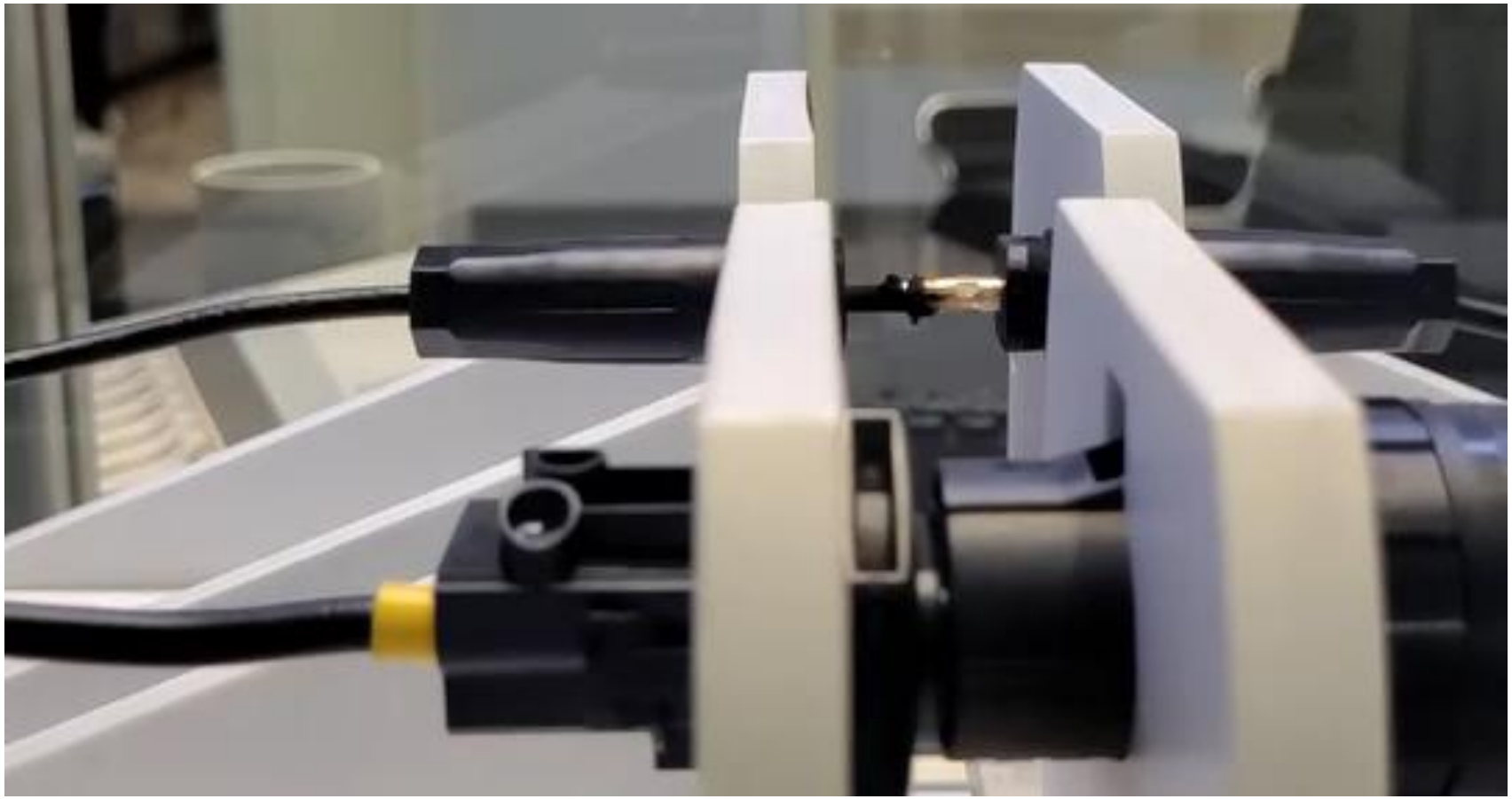
DC Load: 320V, 20A



Arc-free DC Connector (ArcZero 800V, 20A)

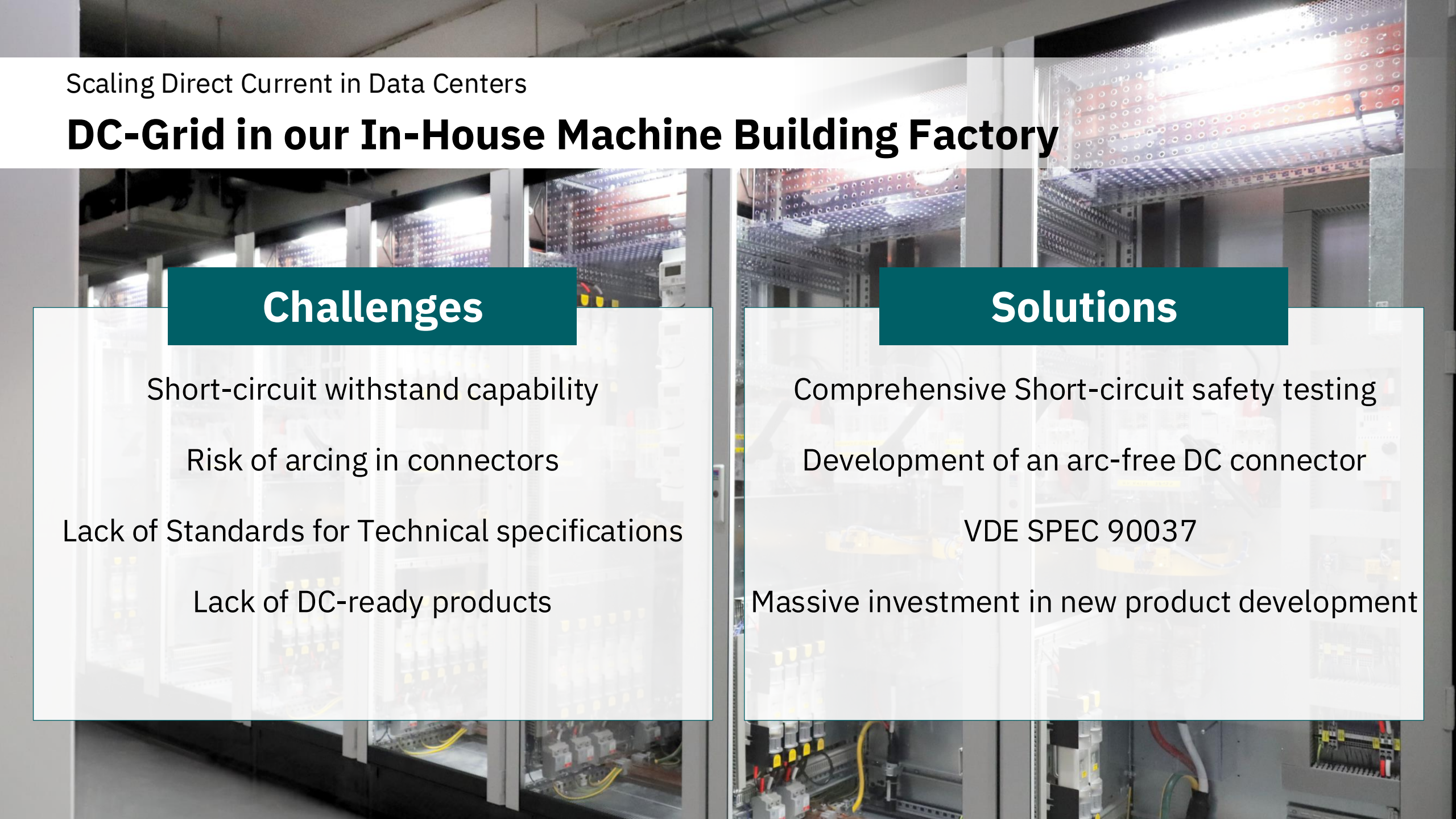
Standard plug

ArcZero



Scaling Direct Current in Data Centers

DC-Grid in our In-House Machine Building Factory



Challenges

Short-circuit withstand capability

Risk of arcing in connectors

Lack of Standards for Technical specifications

Lack of DC-ready products

Solutions

Comprehensive Short-circuit safety testing

Development of an arc-free DC connector

VDE SPEC 90037

Massive investment in new product development

A Core DC-Portfolio is Available and is being Expanded

CONNECT — POWER — SWITCH — PROTECT — MEASURE — CONTROL



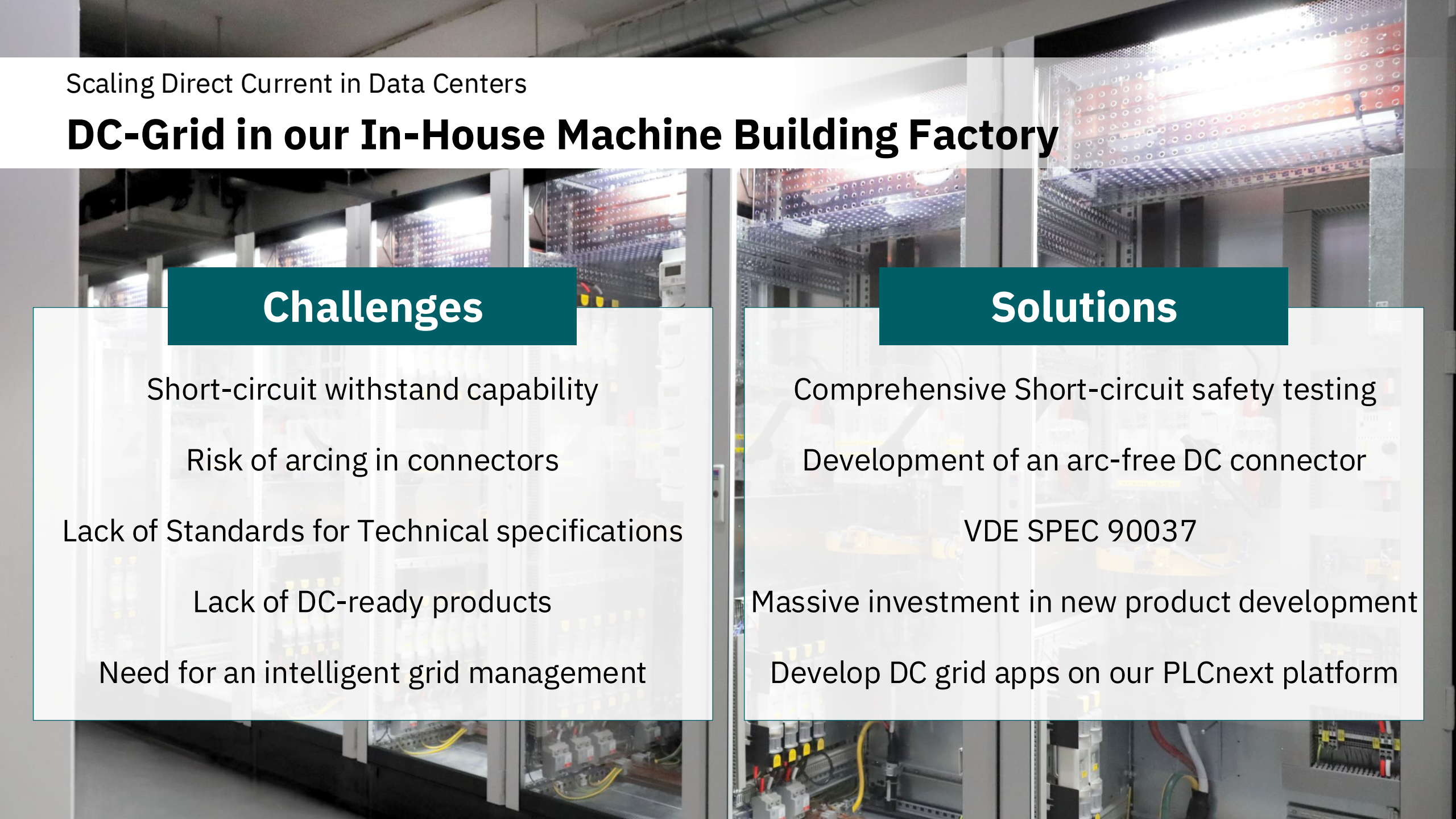





PHOENIX CONTACT in Expo Hall 1, Booth A 13

Scaling Direct Current in Data Centers

DC-Grid in our In-House Machine Building Factory



Challenges

Short-circuit withstand capability

Risk of arcing in connectors

Lack of Standards for Technical specifications

Lack of DC-ready products

Need for an intelligent grid management

Solutions

Comprehensive Short-circuit safety testing

Development of an arc-free DC connector

VDE SPEC 90037

Massive investment in new product development

Develop DC grid apps on our PLCnext platform

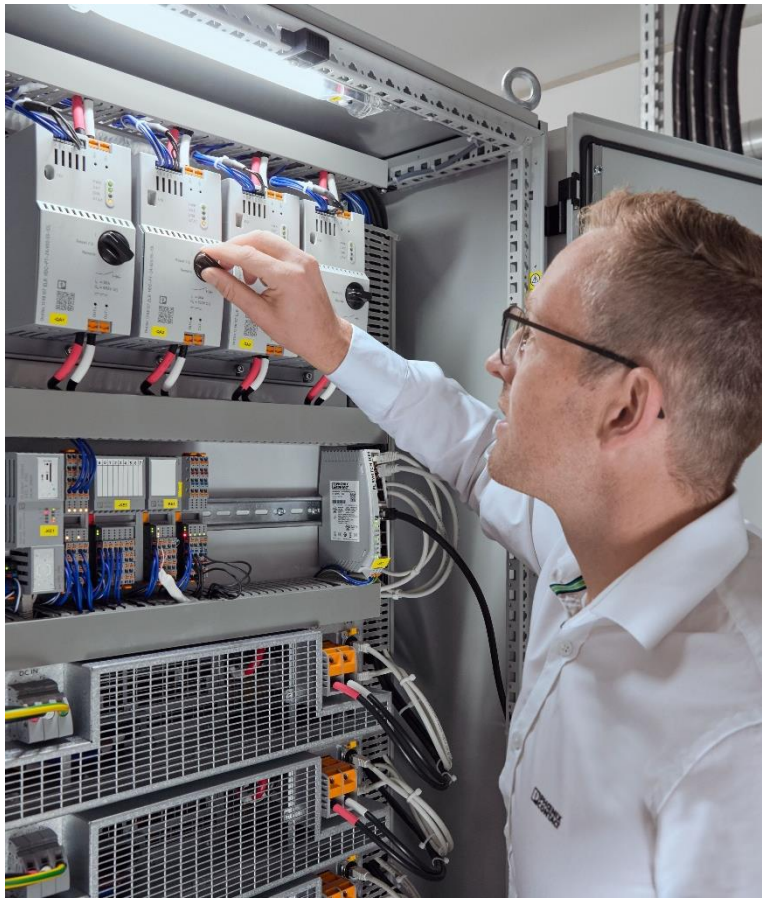
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Data Center

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Why Using DC Grids in Data Centers?

Energy efficiency

Reduce conversion losses

Peak shaving

Reduce load peaks & use breaking energy

Grid stability

Easily integrate energy storage and decrease dependency AC grid

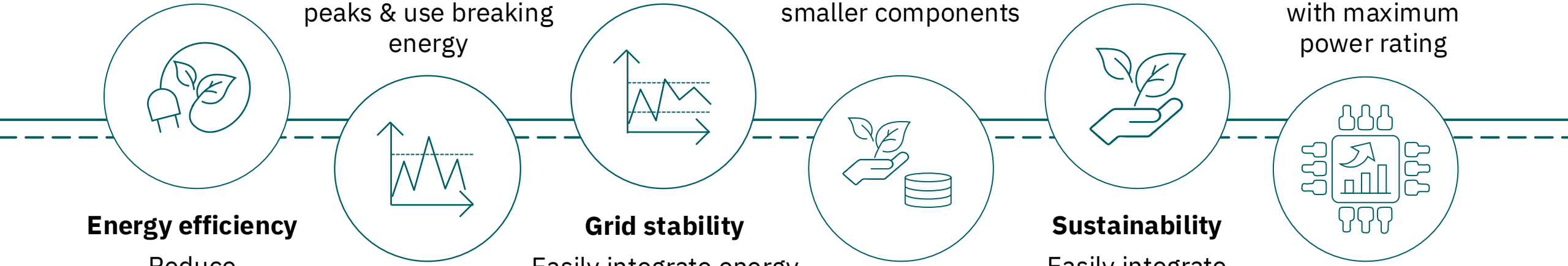
Less resources

Use less copper and smaller components

Sustainability

Easily integrate renewables

Why Using DC Grids in Data Centers?



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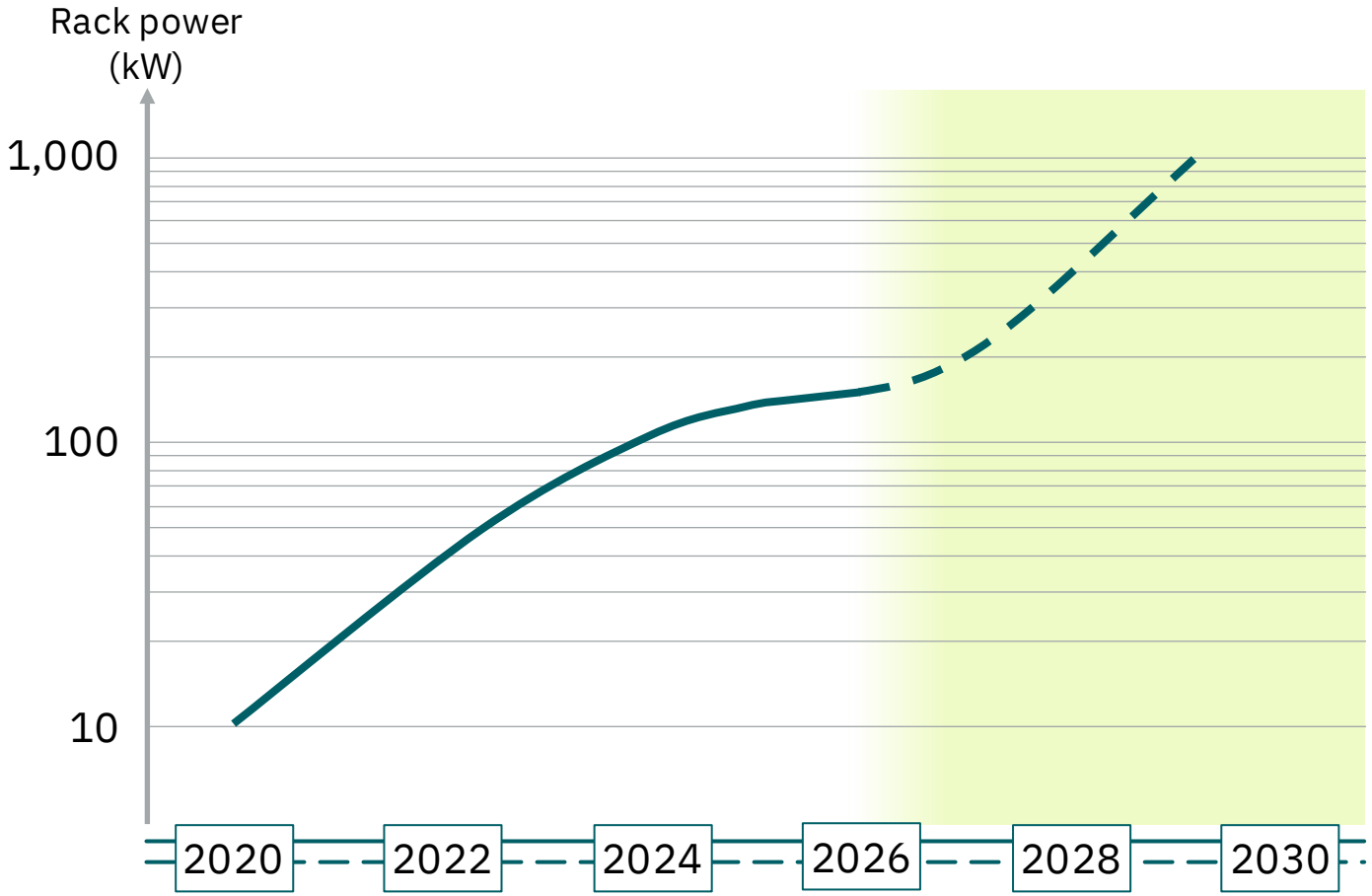
Sustainability

Easily integrate renewables

Power density

Enables server racks with maximum power rating

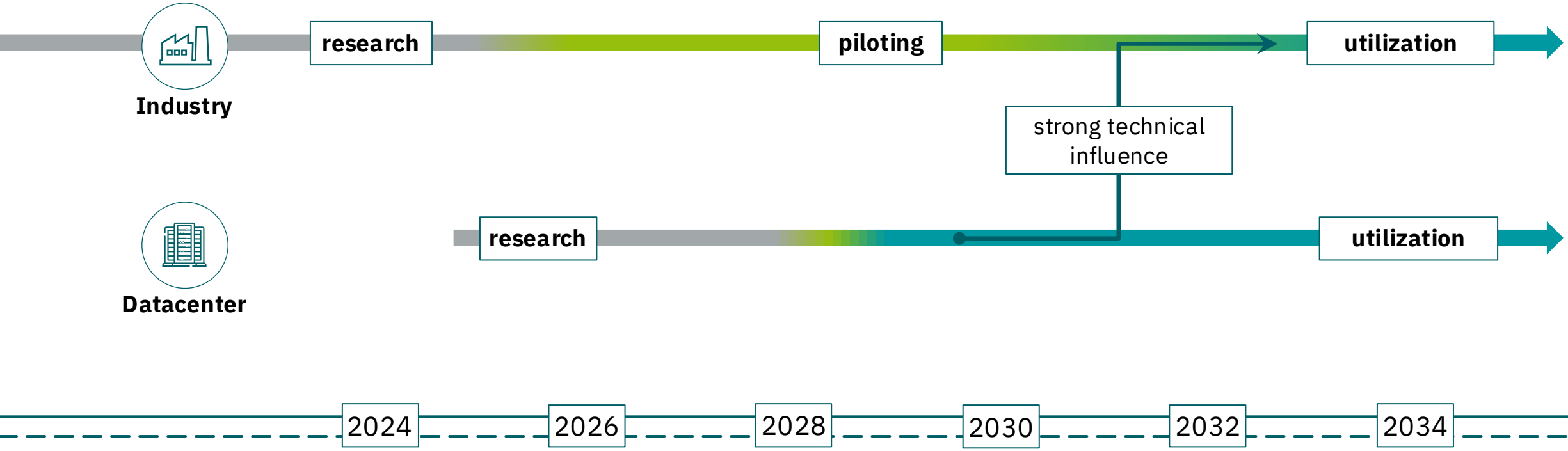
Power Density in Data Center Server Racks is growing fast



Further performance increase only with HVDC

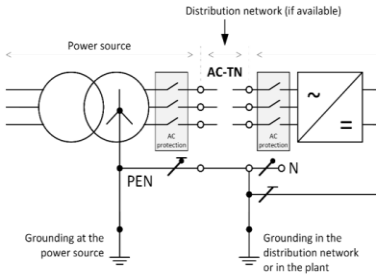
The threshold has been reached **now**!

Direct Current Technology – Estimated Timeframe for Utilization



A wide range of different specifications slows down the adoption

INDUSTRY



± 350 V DC

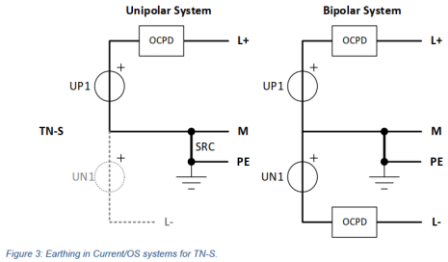
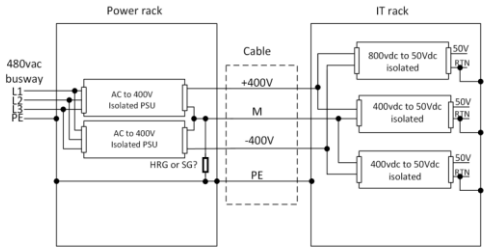


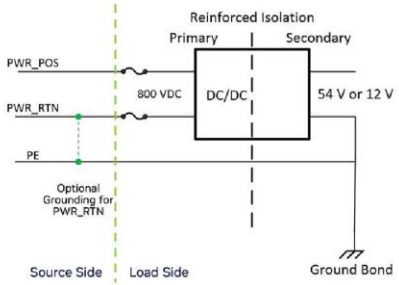
Figure 3: Earthing in Current/OS systems for TN-S.

700 V DC

DATA CENTER

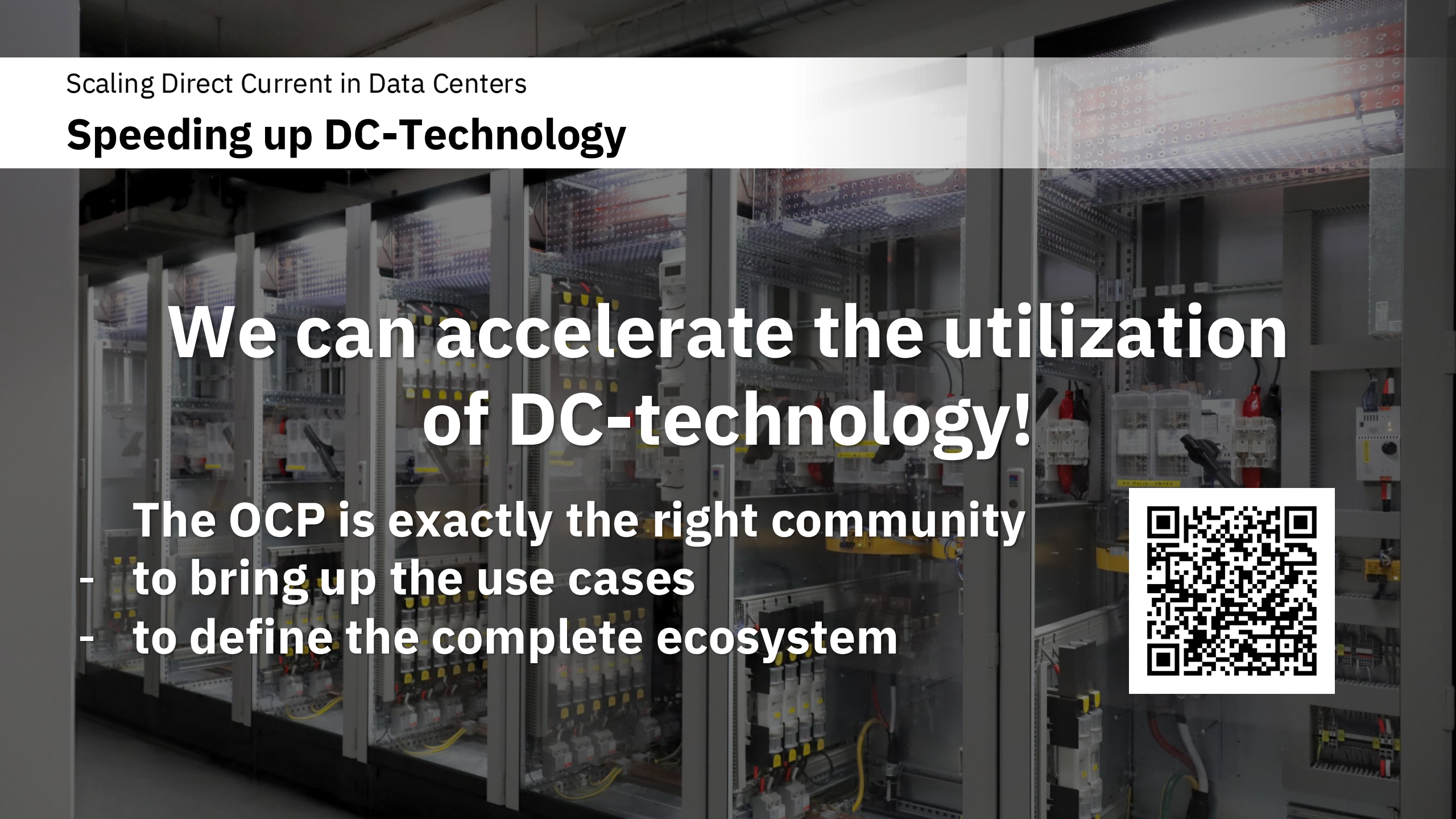


± 400 V DC



800 V DC

Speeding up DC-Technology



We can accelerate the utilization of DC-technology!

The OCP is exactly the right community

- **to bring up the use cases**
- **to define the complete ecosystem**



Thank you

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